Darlyng Rafael Mateo Peral

Software Development Student | Aspiring Software Engineer

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PROFILE

Software Development student with practical project experience in backend development, REST API design, and database-driven systems. Familiar with JavaScript, TypeScript, Go, Kotlin, Python, C++, SQL, and modern software development technologies. Interested in software engineering, algorithms, data structures, and high-performance computing. A self-directed learner interested in understanding how complex software systems are designed, developed, and improved.

SELECTED PROJECTS

Votech Voting Platform

DEVELOPED PROJECT

Voting platform developed for use during a school fair.

- Independently developed the backend using **TypeScript**, **Express.js**, and **Supabase**.
- Designed REST API functionality for managing projects, votes, and results.
- Implemented a custom **JWT**-based authentication system.
- Used during an annual school event with approximately 500 users, without reported service interruptions.

STACK TypeScript · Express.js · Supabase · JWT · REST APIs

CORO Civic Technology Platform

PERSONAL PROJECT · IN DEVELOPMENT

Civic technology platform designed to connect citizens, municipal institutions, and field operations.

- **Backend:** **modular monolith** with **Hexagonal Architecture** in Go (Chi, pgx, sqlc); event-driven processing with Kafka and the Transactional Outbox pattern.
- **Geospatial:** **PostGIS** for geographic queries and **Row-Level Security** for multi-tenant municipal data isolation.
- **Mobile:** two Android apps, a citizen app and CORO Campo, designed offline-first with **Room** and **WorkManager**.
- **Web and security:** Astro landing page, TypeScript management interface, Leaflet maps; Ed25519-signed JWTs, Argon2id hashing, and EXIF metadata removal.

STACK Go · Kotlin · PostgreSQL/PostGIS · Kafka · Jetpack Compose · Room · Astro

Sistema Interno Académico (SIA) Academic Management System

PREVIOUS ACADEMIC PROJECT

Previous academic software project that provided practical experience in backend development and databases.

- Developed backend REST API functionality using **Node.js** and **Express.js**.
- Integrated **PostgreSQL** and **Microsoft SQL Server** databases.
- Implemented **JWT**-based authentication and authorization for academic roles, including administrators, teachers, and students.
- Used **Helmet** for security, **Jest** for testing, and **PDFKit** for PDF document generation.

STACK Node.js · Express.js · PostgreSQL · SQL Server · JWT · Helmet · Jest · PDFKit

TECHNICAL SKILLS

Programming Languages	Go, Kotlin, JavaScript, TypeScript, Python, C++, SQL
Backend Development	Node.js, Express.js, REST APIs, Backend Architecture
Frontend & Mobile	HTML, CSS, React, Astro, Android (Jetpack Compose)
Databases & Data	PostgreSQL, PostGIS, Microsoft SQL Server, Supabase, Room/SQLite, Apache Kafka
Tools & Environment	Git, GitHub, GitHub Actions, Visual Studio Code, Linux, Figma
Testing & Security	Jest, JUnit, MockK, Turbine, JWT, Ed25519, Argon2id, Helmet

EDUCATION

Instituto Politécnico Max Henríquez Ureña (IPMHU) EXPECTED 2028

Technical Program: Application Development and Administration (Desarrollo y Administraciones de Aplicaciones Informáticas)

Santo Domingo, Dominican Republic · Current technical education

CERTIFICATIONS

CertiPlus	Full Stack Expert
INDOTEL	Talento Digital training program, completed

LANGUAGES

Spanish	Native
English	A2 overall (EF SET)

TECHNICAL INTERESTS

Algorithms and Data Structures · Competitive Programming · Software Engineering · High-Performance Computing · Backend Systems · Computer Science